

PENGYU JIE

Software Engineer | Full-Stack Development

Paris, France · +33 7 73 12 40 83 · pengyujie.dev@gmail.com

GitHub: github.com/JayGarland

Portfolio: rajayoux.github.io

LinkedIn: linkedin.com/in/pengyu-jie

PROFESSIONAL SUMMARY

Software engineer with professional experience in healthcare full-stack development and VR client systems, alongside independent cross-platform development across Python/FastAPI, web, and native Android/Kotlin.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, C#, Python, Kotlin, SQL.
Frontend: Angular (17–20), RxJS, HTML5/CSS3.
Backend: ASP.NET Core, FastAPI, Entity Framework Core, REST APIs.
Mobile & Client: Android (Kotlin), Unity.
Testing: xUnit, pytest, JUnit.
Tools & AI: Git, Azure DevOps, CI/CD, Claude Code, Codex, Gemini APIs.
Spoken Languages: Chinese (Native), English (B2), French (B1).

PROFESSIONAL EXPERIENCE

Presage Care — Software Engineer Intern Feb 2026 – Aug 2026 | Paris, France (PariSanté Campus)
Angular · TypeScript · RxJS · C# · ASP.NET Core · EF Core · SQL Server · Azure DevOps

- Production Delivery:** Shipped 88 merged pull requests across three Angular/.NET production codebases covering scheduling, reporting, patient-management, and clinical workflow features for a healthcare platform.
- Multi-Patient Calendar:** Developed a full-stack calendar for care coordinators, aggregating multiple clinical event sources in ASP.NET Core and discarding stale responses during rapid patient and date navigation in Angular.
- Weekly Brief:** Delivered weekly facility health reporting for care-home directors, aggregating health and intervention metrics in ASP.NET Core for an Angular dashboard while keeping scheduled email rendering separate.

MangoTV — VR Applications Intern Dec 2023 – Apr 2024 | Changsha, China
Unity · C# · VR/XR

- VR Client Development:** Contributed to a multiplayer Unity/C# VR application, working across client features, networking integration, runtime asset loading, and headset adaptation.

SELECTED PROJECTS

Resonova — Cross-Platform Audio Application Mar 2026 – Present
Python · FastAPI · TypeScript · Kotlin · Android · Spotify API · Gemini API

- Android Playback:** Developed Resonova's native Kotlin playback layer to coordinate Spotify tracks with generated commentary, including background playback, audio focus handling, and lock-screen media controls.
- Playlist Sequencing:** Designed and implemented a Python sequencing algorithm that adapts track selection and placement from listening history, skip patterns, replays, and position-specific constraints.
- AI Audio Generation:** Integrated Gemini-generated dialogue, speech synthesis, and Spotify tracks in a FastAPI pipeline that produces personalized audio episodes.

Dashcam Video Integrity Validator Jan 2026 – Apr 2026
C# · .NET · FFmpeg · Perceptual Hashing

- Video Integrity Verification:** Implemented a C#/.NET validator that compares uploaded video against perceptual hashes recorded at capture time, using temporal matching to detect timestamp drift and missing segments while tolerating re-encoding.

EDUCATION

ESIGELEC Sep 2024 – Sep 2026 | Rouen, France
Master of Science in Software Engineering & Digital Transformation
Coursework and internship defense completed; degree conferral pending formal validation. GPA: 16.45/20 (Rank 2/16). Youth Achievement Merit Scholarship.

Hainan Vocational University of Science and Technology Sep 2020 – Jun 2024 | Haikou, China
Bachelor of Science in Software Engineering
GPA: 3.7/4.0 (Rank: 20/443).

HONORS & AWARDS

Malaysia Technology Expo (MTE 2023) Gold Medal & Special Award (Outstanding Innovation)
China Collegiate Computing Contest (2023) Provincial First Prize
China Collegiate Computing Contest (2022) National Second Prize